

Figures: Branch architectures, fact-checking module, and overall system

The figures below illustrate the three modality-specific detection branches, the fact-checking module, the overall fusion architecture, and the training procedure used to build the system.

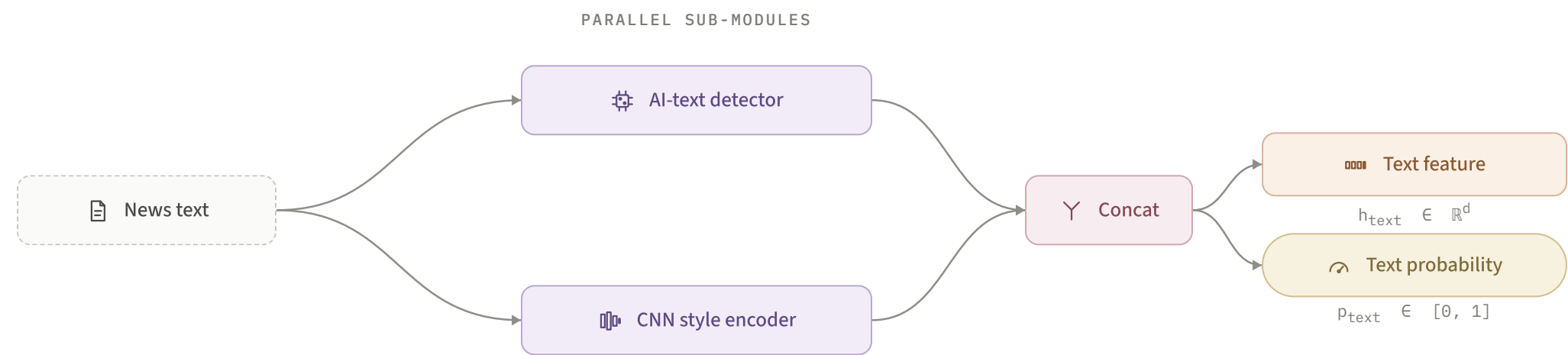


Figure 1. *Text Credibility Branch.* The news text is processed by two parallel sub-modules — an AI-text detector that estimates the likelihood the article was machine-generated, and a CNN-based style encoder that captures lexical and stylistic cues. Their outputs are concatenated to form a single text feature vector and an aggregated text-credibility probability.

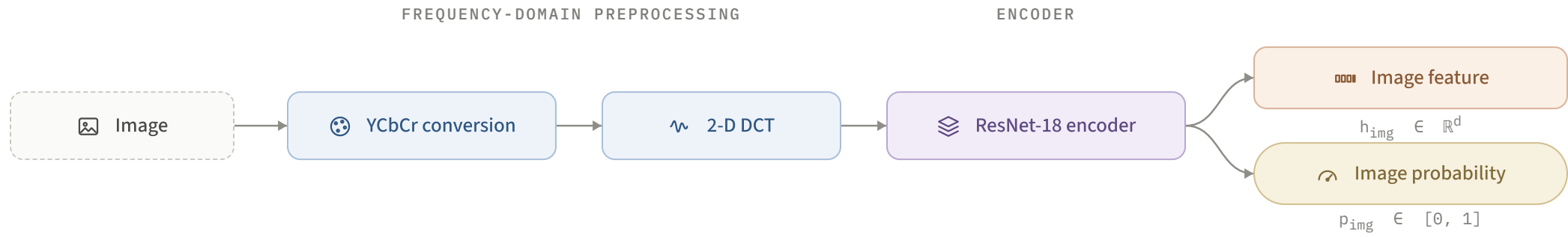


Figure 2. *Image Authenticity Branch.* The input image is converted from RGB to YCbCr, decomposed into the frequency domain via a 2-D Discrete Cosine Transform, and fed to a ResNet-18 encoder. The encoder yields an image feature vector together with an authenticity probability.



Figure 3. *Cross-Modal Consistency Branch (FND-CLIP).* Text and image inputs are jointly embedded by FND-CLIP, which produces a cross-modal feature vector and a consistency probability quantifying the semantic alignment between the two modalities.

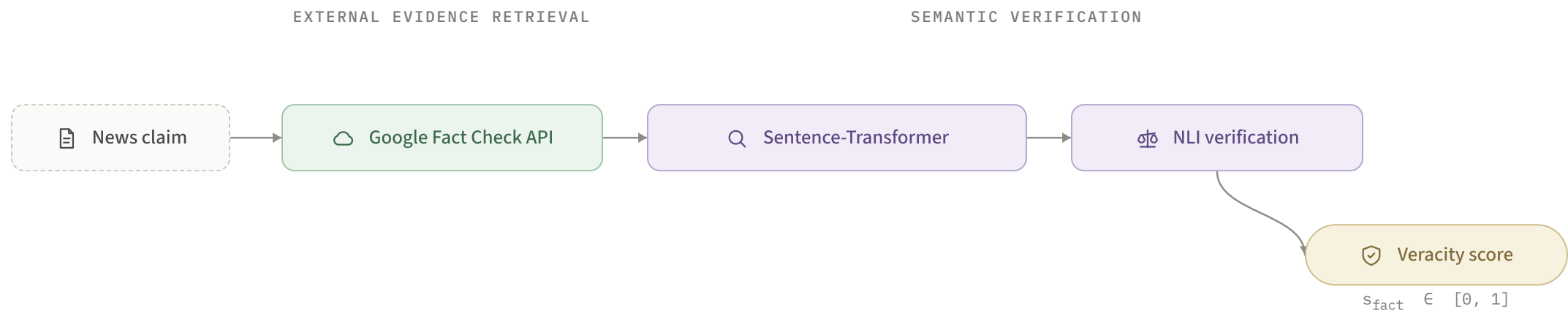


Figure 4. *Fact-Checking Module.* A news claim is queried against the Google Fact Check API; candidate evidence is retrieved by a Sentence-Transformer semantic search and then verified by a Natural Language Inference model that contrasts the claim with the evidence to produce a soft veracity score.

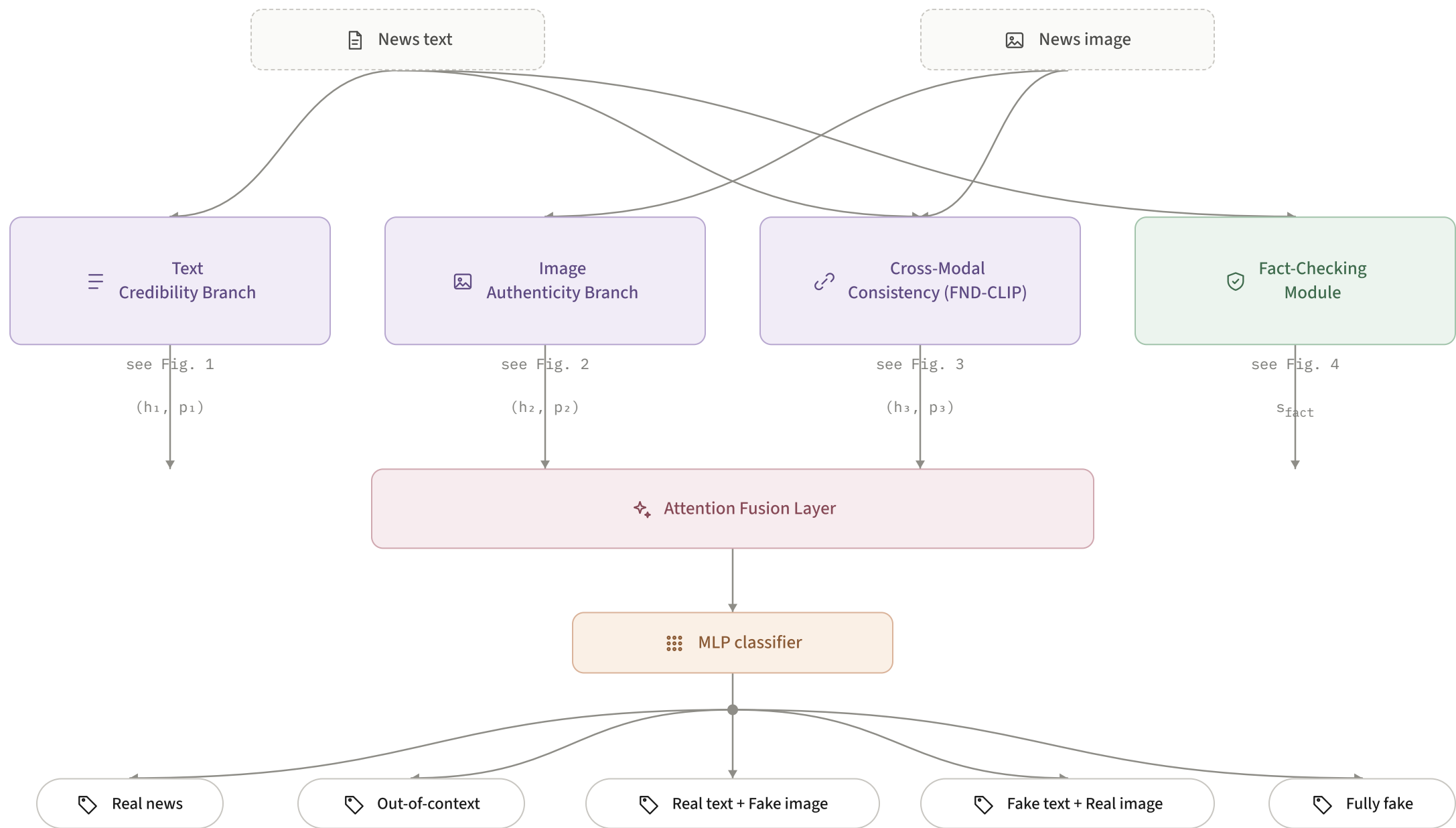


Figure 5. Overall System Architecture. The text and image of a news article are dispatched to four parallel modules: the three modality-specific detection branches and the fact-checking module. Each module emits a feature vector and a calibrated probability (or soft score). The four signals are combined by an Attention Fusion Layer that learns to weight branches per sample and then passed to an MLP classifier, which assigns one of five misinformation categories.

